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Systematic Literature Review: An Investigation Towards Finding Constructs For Performance Prediction of Students in an Online Engineering Course

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***Abstract*—The use of technology in the field of engineering education has been the most common intervention, especially in the post-pandemic era. Most universities have plans to continue the blended approach to education in the future. This decision has to be an evaluated decision and the analysis of a student's performance serves as an input to take the decision. The other advantage of analysis include early prediction of student performance which can help the instructors to provide timely interventions and help the students to improve their performance. Thus identification of constructs that reflect student engagement and performance in a course delivered in online mode is very essential. This literature review attempts to bring forward the constructs used by various researchers that reflect student engagement and performance. The review is situated in the context of engineering education delivered in online mode. The identification of constructs is significant and helps to build machine learning models for predicting the performance of the students. Standard Systematic Literature Review(SLR) methods defined in literature including citation searching and hand searching were carried out to identify the constructs that have been in use. A list of constructs used by researchers in the literature, that capture students' attention and performance are identified and presented in this review. The identified constructs include students' interaction with content, students' interaction with peers, demographic factors, and the academic records of the student. These validated constructs are proposed to integrate with the Learning Management System(LMS) and use the feature for early prediction of student failures.**

Keywords*—*Systematic Literature Review, Student engagement, Blended learning, Learning Management System

I. INTRODUCTION

The blended mode of delivery in the field of education has become quite prominent during this pandemic era [1], [2]. The decision to continue the usage of blended mode must be evaluated. Students' academic performance can be a parameter that can help educators to make decisions towards continuing the blended mode of delivery. Thus identification of constructs that reflect students' engagement and performance in blended mode becomes more essential.

The literature presents several studies that focus on describing the impact of different factors on a student's academic performance in the context of Indian engineering education [1], [3]–[5]. Further, the complexity of the study increases when other dimensions such as blended and online modes of delivery get added.

Performance of the student depends on how well the students are provided with an opportunity to have interaction with content, peers, and mentors [6]. Hence there is a need for operationalizing the action 'interaction' in the context of online education. Through this paper, the authors attempt to bring forward the constructs used by various researchers in the literature that have influenced student engagement and performance in the blended/online mode of education.

Online mode of education has become quite prominent in recent years. Platforms like Learning Management Systems(LMS), Massive Open Online Courses(MOOCs), and other Virtual Learning Environments(VLE) are largely used for online delivery. The amount of interaction and the kind of interaction that the students have with such environments have been proven influence the students' performance [7]–[13].

Apart from the impact of the interaction of students with content on students' performance, some other factors have also been studied intensively by the researchers. A few studies tell us about the impact of the interaction of a student with peers on the academic performance of the student [14]. The activities of the student on the discussion forums and social media platforms are considered under peer interaction [11]–[13], [15], [16]. A few authors also tell us about how demographic factors affect students' performance [5], [7], [17], [18].

The motive of this review is to identify the constructs currently used by the researchers that reflect student engagement and performance in the engineering courses delivered in blended mode.

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The rest of the paper is organized as follows. Identification of methodologies adopted by researchers when approaching the problem of performance prediction is reviewed in Section II. The methodology adopted is described in Section III. The analysis of the factors identified in section 2 is reviewed in Section IV. Section V concludes the study and provides some directions for future work.

II. Discussion of Methodologies Adopted by Researchers

This review has collected and synthesized the existing works and reviews report (publications from journals conferences, invited papers, and textbooks) and identified the factors and methods used when predicting students' performance and also the evaluation metrics used by them when assessing the students [19]. The following section discusses the same in detail.

A. Factors

The factors considered to predict the performance of a student in most of the literature can be broadly classified into the following categories: content interaction, peer interaction, academic records, and demographic factors [6]. Content interaction includes the activities of the student on online educational platforms like Learning Management Systems (LMS) [7], Massive Open Online Courses(MOOC) [9], etc. Activities of the student on the discussion forums and social media platforms are considered to understand the student's interaction with peers [11], [13], [15], [16]. The previous academic records and formative assessment marks are included under the academic records [14], [20]. The domestic factors(parental information, socioeconomic status, etc.), and school-related factors(geographic location of the school, school quality, etc.) are considered under the demographic factors [7], [8], [17]. Some works also propose the usage of multimodal data to understand the learner's behavior as well [21].

Content interaction, peer interaction, and academic records also form three of the four main pillars of the LMS [22]. Online platforms should be seen as a tool to increase student involvement through self-reflection rather than just a place to dump the content. Ensuring the employment of the above-mentioned pillars and proper administration, which is the fourth pillar of LMS, will increase the chances of the students succeeding academically.

B. Methods

Various kinds of classification algorithms have been used by researchers to predict the performance of a student. Algorithms based on machine learning (ML) and deep learning (DL) are the state of the art that helps the educator in multiple ways. The capabilities of these ML and DL algorithms to handle massive amounts of data, and data from multi-modalities made these algorithms the state of art [23]. They are adept at managing multi-dimensional data. As the algorithms gain experience, their efficiency and accuracy continue to increase.

Machine learning (ML) methods like the random forest, decision tree, and support vector machine [8], [11], [15], [18] have often been used to predict students' performance. Deep learning algorithms like Artificial neural networks (ANNs) and long short-term memory (LSTM) [7], [14], [20], [24]have been used by some academicians and researchers to forecast students' achievements. The danger of over-fitting with limited data is larger for DL methods, which are often more complicated [25]. Hence, ML algorithms have been used in cases where adequate data is unavailable.

C. Performance Evaluation Metrics

Most works have classified the students' performances into pass/fail/distinction based on their performance [7], [10], [17], [26]. Some of the remaining papers have predicted the performance of the student using regressors and have performed regression tasks [18].

III. Methodology

The current study adopted the steps mentioned in the SLR [27] process to carry out the literature review. In this study, the works that were published between 2010 to 2021 were considered. This used keywords and phrases like performance prediction, blended learning, and online learning, in search engines like Google scholar. Along with this, citation searching was used to find relevant literature. Citation searching refers to searching for literature cited by the sources of already referred papers as well as the sources cited by them [27]. The keywords identified here were performance prediction, learner engagement, clickstream data, peer interaction, demographic data, and course performance.

The motivation to perform the SLR was to identify the constructs commonly used in predicting students' performance. These validated constructs are proposed to integrate with the Learning Management System (LMS) of a university that was in need of evaluating the effectiveness of the newly adopted blended mode of delivery. The data(academic records of the students, the activities of the students on the online platform, etc) obtained regarding the students of the university was mostly quantitative, and hence this study primarily focused on the works of researchers who have conducted the quantitative analysis.

The information from the SLR is synthesized and represented through a literature map [28] as shown in figure 1. The literature map is a two dimensional graphical representation method that helps to organize the learning. It has 'nodes' and 'links'. Nodes represent the 'lens' or 'main theme' and the references show the source of information. From the entire collection, the relevant articles were identified and further studied for this review. Figure 2 represents the PRISMA flow diagram [29] of the literature review conducted. The flow of this diagram depicts the flow of information through the different phases of a systematic review [27]. It maps out the number of records identified, included, and excluded, and the reasons for exclusions [30].

Performance prediction has been studied extensively in the past for various purposes like identifying the students

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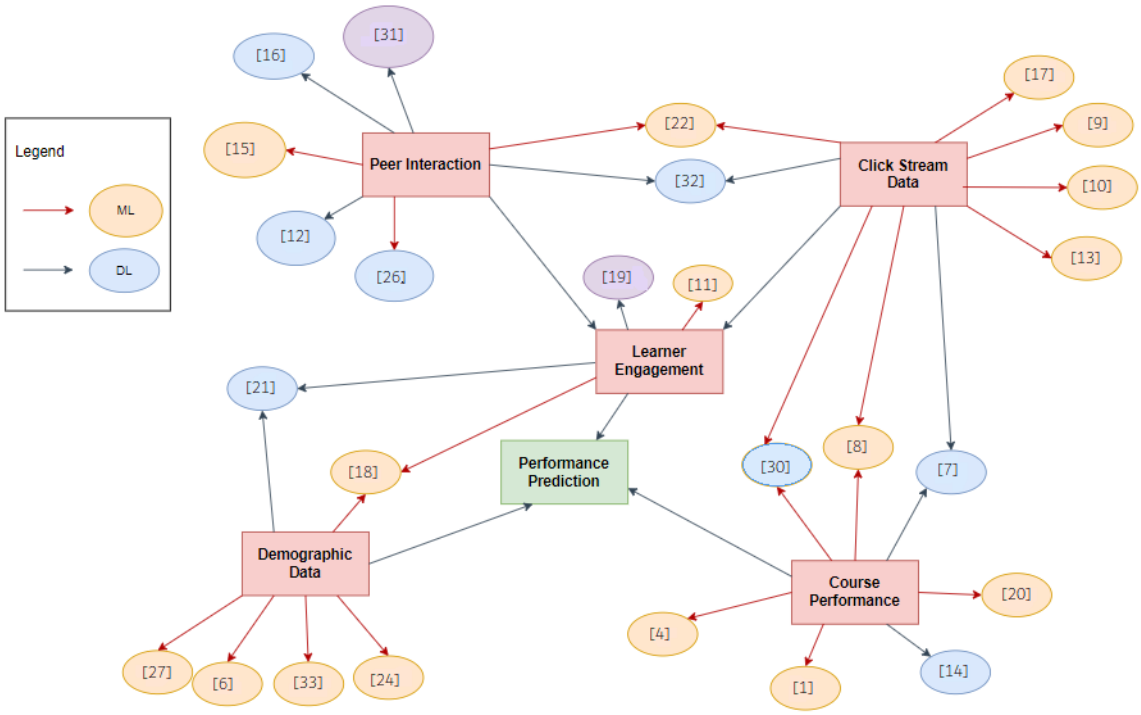

Fig. 1: Literature Map

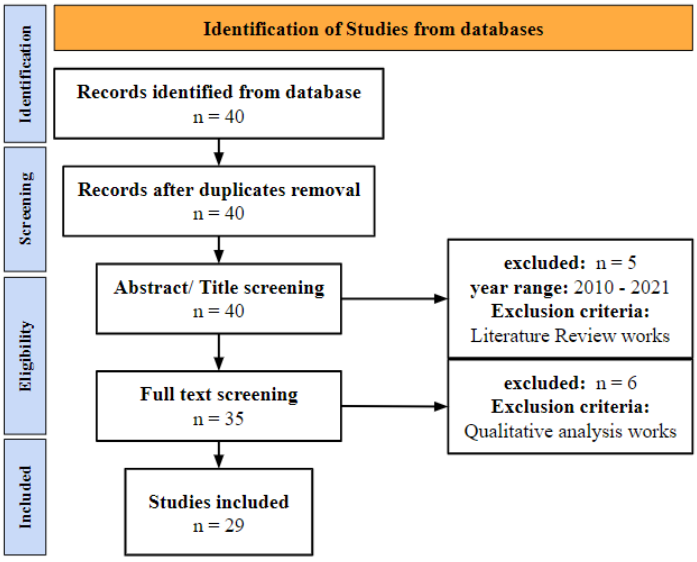

Fig. 2: PRISMA Flow Diagram

at risk, identifying and customizing the student learning environments, etc. The predicted values are the grade point averages (GPA), knowledge, scores, etc. Most works have used quantitative approaches and classification algorithms to classify the students into various categorical values (for example Pass/Fail).

The study limited the literature search to papers that exclusively dealt with quantitative analysis. This review looked at the aspects that these works considered and how they influenced students' performance. The researchers' methodologies and the models they trained and evaluated for prediction were also taken into consideration. The expected output was another factor that was noted from all of the research works considered.

The performance of a student is influenced by various factors. Learner engagement [9], course performance details [14] and the demographic data [17], [18] are a few of the factors considered and reviewed in our study. Learner engagement here is further categorized into peer interaction data and clickstream data, which indicates the content interactions.

The factors thus identified and obtained are further used

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to predict students' performance. Several machine learning and deep learning methods are used to perform the task of prediction as mentioned in the previous section. The research works that have made use of ML and DL methods are depicted using different colours in the literature map in figure 1. Machine learning-based methods are marked with orange colour, whereas deep learning-based methods are shown in blue colour in the literature map. The literature map also has a few purple-coloured ovals, they include qualitative analysis works and literature survey papers.

IV. Analysis

The literature review aimed to identify the constructs currently used by the researchers that reflect student engagement and performance. The factors, methods, and predicted outputs have been noted as a part of this review.

The factors discovered in the literature can be broadly divided into four categories namely- demographic characteristics, academic records, content interaction, and peer interaction [13]. A complete discussion of these factors may be found in Table 1. In the recent past factors like the number of clicks per day, number of active days, number of resources accessed, number of videos watched and activities of students on discussion forums were commonly used and have proven to provide excellent results in predicting the performance [7], [15], [31].

TABLE I: Attributes commonly used

Identified Category	Attributes Considered	References
Demographic Factors	Gender, age, parent's educational level, parents' involvement, students study environment, school and classroom environment, possession of computers, region, family income and expenditure, nationality, birthplace.	[7], [8], [10], [11], [17], [18]
Academic records	Internal assessment marks, assignment submission marks, past performances.	[10], [14]
Content interaction	Click stream data, number of clicks per day, number of active days, number of resources accessed, number of videos watched, announcement views.	[7]–[11], [17], [31], [32].
Peer interaction	Activities of students on wiki, blog posts and twitter or any other discussion forums.	[11], [12], [15], [16], [33]

The algorithms utilized were divided into two categories: machine learning and deep learning. Table 2 lists the various Machine Learning and Deep Learning algorithms that were used in this study. In general, classification approaches are seen to be employed more frequently. Random forest, support vector machine, and decision trees are the most commonly used classification algorithms giving an accuracy of over 95% [7], [8], [11], [18], [34].

Various kinds of metrics were used to describe the

TABLE II: Methods Commonly Used

Methods	Approaches	References
Machine Learning	Logistic Regression, SVM, Naive Bayes, Decision Tree, Random Forest , XGBboost.	[8], [11], [15], [18]
Deep Learning	Deep Neural Networks(DNN), Long short term memory(LSTM),etc.	[7], [12], [31], [32].

students' performance. Mostly they were classified into pass/fail/distinction or other such categories using multi-class classification algorithms [7], [10], [17], [26]. These classifications were made on based on their final results (grade point averages or the total final marks).

V. Conclusion

This paper summarizes the literature and identifies the constructs that reflect student engagement and performance. The identified constructs that reflect student engagement and performance include demographic factors, academic records, and the factors dealing with the content level of interaction and peer interactions. In this post-pandemic era, the inclination toward the online and blended mode of delivery has increased. Hence, the student's interaction with content and peers mostly constitute the data obtained from online platforms. This paper not only summarizes the factors influencing students' performance but also discusses the methods used and the evaluation metrics used to evaluate the student's performance considered by the researchers.

This review primarily focuses on the research works that have used quantitative analysis methods, future works can take into account the research works that have used qualitative analysis techniques and those works that integrate both qualitative and quantitative methods. Additionally, studies that have employed clustering and statistical methods like regression can be taken into account and reviewed.

The understanding developed after reviewing the current trends and the conclusions provided by other researchers in the field of students' performance prediction and artificial intelligence helped the first author further their research. The data for the study collected included click stream data from Learning Management System(LMS) to understand the students' interaction with the content and collaboration made by the student on the GitHub platform to understand the students' interaction with their peers. The other sets of data included students' academic performance and demographical data.

This data was further used to train the classification algorithms. The machine learning based classifications algorithms like Random forest, Naive Bayes, Decision tree, Support Vector Machine, and XGBoost were used to predict the performance [35]. And multiclass classification was done using the decision tree classifier achieving a classification accuracy of 96%. The students were grouped into four categories including 'Excellent', 'Good', 'Average', and 'Poor'. The authors are further working towards integrating these identified constructs with the LMS at

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the university to create a personalized learning for the students.

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